Multi-Symbol Discrete Stationary Information Source

Luping, Xiang (向路平)

luping.xiang@nju.edu.cn

Office: 南雍楼西231

Main Content

1. Multi-Symbol Information Source
2. Memoryless Information Source
3. Memory Information Source

Main Content

1. Multi-Symbol Information Source
2. Memoryless Information Source
3. Memory Information Source

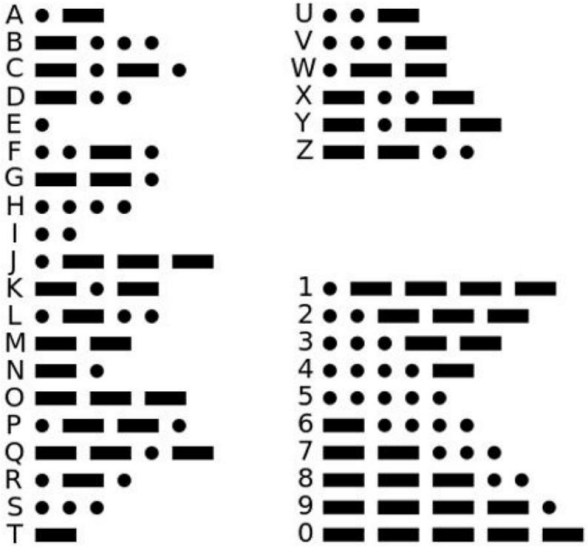
Multi-Symbol Discrete Info. Source

- An information source normally outputs a sequence of random symbols. The symbols are statistically correlated with each other. The sequence of these random symbols can be expressed as

$$\boldsymbol{X} = X_1,\ X_2, X_3,\ \cdots$$

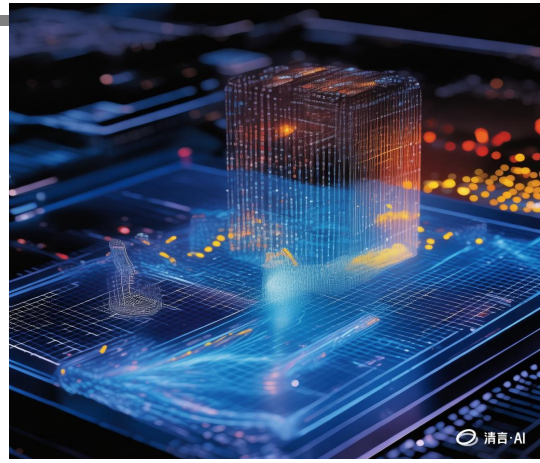
In a generic case, the random symbols X_i and X_{i+r} at different moments have different statistical properties.

- Typical multi-symbol discrete information source:
  - English Language: Hello Word
  - Morse Code: .... . .-.. .-.. --- / .-- --- .-. .-.. -..
  - *The first official message was sent by Samuel F.B. Morse on May 24, 1844, to open the Baltimore-Washington telegraph line.*

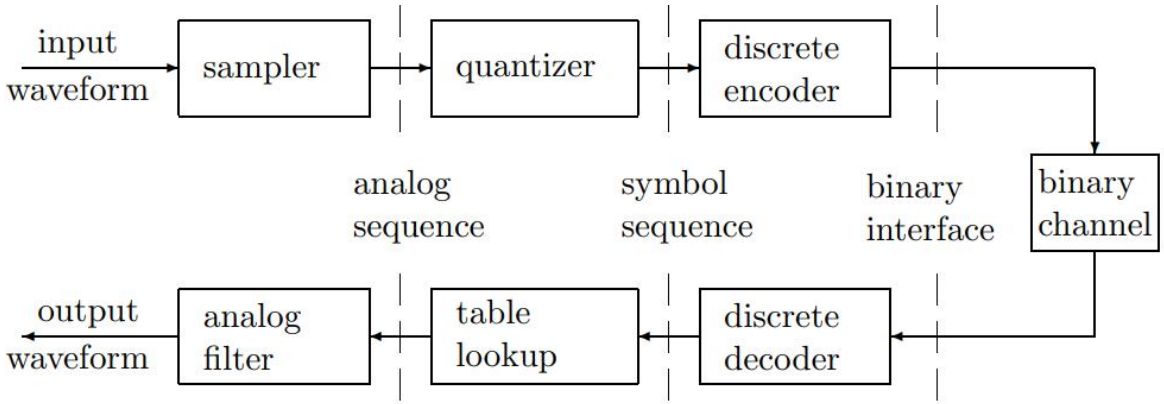


Multi-Symbol Discrete Info. Source

Multi-Symbol Discrete Info. Source



- **What kind of information source is called stationary source?**
- Suppose that in the random vector $\boldsymbol{X}$, the joint probability of the random variables of any dimension does not change as the time goes. *The probability distribution is not related to the time*.
- This is the definition of multi-symbol discrete stationary information source



1D and 2D Stationary Info. Source

- **One-dimensional stationary information source**: given a sequence of random variables $\boldsymbol{X} = X_1 X_2 \cdots X_N$, the random symbols sent at any moment have identical probability distribution:

$$P(X_i = a_1) = P\left(X_j = a_1\right) = p(a_1)$$
$$P(X_i = a_2) = P\left(X_j = a_2\right) = p(a_2)$$
$$\cdots$$
$$P(X_i = a_n) = P\left(X_j = a_n\right) = p(a_n)$$

This source sends symbols with probabilities of $\{p(a_1), \cdots, p(a_n)\}$ at any moment.

$$a \to a \to b \to a \to c \to b$$

- **Two-dimensional stationary information source:** Both the one- and two-dimensional probabilities of the source are unrelated to the time.

$$P(X_i X_{i+1}) = P\left(X_j X_{j+1}\right),\ \forall i,\ j \in Z,\ i \neq j$$

$$\begin{pmatrix} a \\ b \end{pmatrix} \to \begin{pmatrix} a \\ c \end{pmatrix} \to \begin{pmatrix} b \\ b \end{pmatrix} \to \begin{pmatrix} a \\ a \end{pmatrix} \to \begin{pmatrix} c \\ a \end{pmatrix}$$

Discrete Stationary Info. Source

- If any dimensional probability distribution of the discrete information source $\boldsymbol{X}$ is unrelated to the time, for an arbitrary pairs of moments i and j, we have the following equalities:

$$P(X_i) = P(X_j),$$
$$P(X_i X_{i+1}) = P(X_j X_{j+1}),$$
$$P(X_i X_{i+1} X_{i+2}) = P(X_j X_{j+1} X_{j+2}),$$
$$\cdots$$
$$P(X_i X_{i+1} X_{i+2} \cdots X_{i+N}) = P(X_j X_{j+1} X_{j+2} \cdots X_{j+N}),$$

The probability distribution of a single symbol generated at the moment *i* is the same as the symbol generated at the moment *j*.

This is the discrete stationary information source.

Definition in textbook

A stochastic process is said to be stationary i if the joint distribution of any subset of the sequence of random variables is invariant with respect to shifts in the time index;that is,

$$\Pr\{X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\}$$
$$= \Pr\{X_{1+l} = x_1, X_{2+l} = x_2, \ldots, X_{n+l} = x_n\}$$

for every n and every shift l and for all x1,X2,...,Xn∈X.

Mathematical model

- The mathematical model of a **single-symbol** discrete info. source is:

$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} a_1, & a_2, & \cdots & a_n \\ p(a_1), & p(a_2), & \cdots & p(a_n) \end{pmatrix}, \sum_{i=1}^{n} p(a_i) = 1$$

- A **multi-symbol** discrete info. source is $\boldsymbol{X} = X_1 X_2 \cdots X_N$, where we have $X_1, X_2, \cdots, X_N \in \{a_1,\ a_2, \cdots, a_n\}$. Its mathematical model is:

$$\begin{pmatrix} \boldsymbol{X} \\ P(\boldsymbol{X}) \end{pmatrix} = \begin{pmatrix} \alpha_1, & \alpha_2, & \cdots & \alpha_{n^N} \\ p(\alpha_1), & p(\alpha_2), & \cdots & p(\alpha_{n^N}) \end{pmatrix},$$

where we have $\alpha_i = (a_{i_1}, \cdots, a_{i_N})$, $i = 1, \cdots, n^N$; $i_1, \cdots, i_N = 1, \cdots, n$. The probability of a multi-symbol message α_i being sent is $p(\alpha_i) = p(a_{i_1} a_{i_2} \cdots a_{i_N})$. All the probabilities satisfy the following constraint:

$$\sum_{i=1}^{n^N} p(\alpha_i) = \sum_{i_1=1}^{n} \sum_{i_2=1}^{n} \cdots \sum_{i_N=1}^{n} p(a_{i_1} a_{i_2} \cdots a_{i_N}) = 1$$

Main Content

1. Multi-Symbol Information Source
2. **Memoryless Information Source**
3. Memory Information Source

Memoryless Information Source

- For an arbitrary moment i, if the probability of a multi-symbol message generated by a stationary discrete information source satisfy the following equality:

$$P(X_i X_{i+1} \cdots X_{i+m}) = P(X_i)P(X_{i+1})\cdots P(X_{i+m}),$$

  it indicates the independence among the symbols in this message. This is the memoryless information source.
- $X \in \{0,1\}$ is a single-symbol binary information source. Taking a pair of symbols independently generated by X as a message, we have a discrete stationary memoryless info source $\boldsymbol{X} = X^2 \in \{00,\ 01,\ 10,\ 11\}$, regarded as the expanded info source of X having an order of 2.
- Taking three symbols independently generated by X as a message, we have a discrete stationary memoryless info source $\boldsymbol{X} = X^3 \in \{000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111\}$, which is regarded as the expanded info source of X having an order of 3.

Expanded Source of Order *N*

- The mathematical model of a single-symbol discrete info. source is:

$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} a_1, & a_2, & \cdots & a_n \\ p(a_1), & p(a_2), & \cdots & p(a_n) \end{pmatrix}, \sum_{i=1}^{n} p(a_i) = 1$$

- The mathematical model of its expanded memoryless source X^N having an order of N is :

$$\begin{pmatrix} X^N \\ P(X^N) \end{pmatrix} = \begin{pmatrix} \alpha_1, & \alpha_2, & \cdots & \alpha_{n^N} \\ p(\alpha_1), & p(\alpha_2), & \cdots & p(\alpha_{n^N}) \end{pmatrix},$$

  where every message α_i having N independent symbols generated by X and $p(\alpha_i)$ is the probability of α_i being sent.

- Since X^N is memoryless, the probability of the multi-symbol message $\alpha_i = (a_{i_1}, \cdots, a_{i_N})$ being sent is expressed as

$$p(\alpha_i) = p(a_{i_1})p(a_{i_2})\cdots p(a_{i_N}), i_1, \cdots, i_N = 1, \cdots, n$$

Example

- We have a single-symbol source $X = \begin{Bmatrix} 0 & 1 \\ 1/4 & 3/4 \end{Bmatrix}$.
  - Please calculate the entropy of the information source X;
  - A multi-symbol source $\boldsymbol{X}$ sends messages having 100 symbols independently generated by X . Please calculate the entropy of the multi-symbol source $\boldsymbol{X}$.

Entropy of the expanded info. source

- The entropy of the expanded info. source $\boldsymbol{X} = X^N$ of order N is

$$H(\boldsymbol{X}) = H(X^N) = -\sum_{X^N} P(\boldsymbol{X}) \log_2 P(\boldsymbol{X})$$

$$= -\sum_{X^N} p(\alpha_i) \log_2 p(\alpha_i)$$

- The entropy of the expanded info. Source and that of the single-symbol info source satisfy the equality of $H(X^N) = NH(X)$.

  Proof: Exploit the entropy definition and the memoryless property of the expanded information source.
- The entropy of the expanded source can be directly obtained by its corresponding single-symbol source.
- The entropy of the expanded info source is the average information carried by the messages having multi-symbols.

Main Content

1. Multi-Symbol Information Source
2. Memoryless Information Source
3. **Memory Information Source**

Discrete Memory Info. Source

- The symbols sent by the discrete stationary memory information source are statistically correlated with one another. This statistical correlation can be represented in the following two manners:
  - For the N-dimensional discrete stationary memory information source $\boldsymbol{X}$

$$\begin{pmatrix} \boldsymbol{X} \\ P(\boldsymbol{X}) \end{pmatrix} = \begin{pmatrix} \alpha_1, & \alpha_2, & \cdots & \alpha_{n^N} \\ p(\alpha_1), & p(\alpha_2), & \cdots & p(\alpha_{n^N}) \end{pmatrix},$$

    Where $\alpha_i = (a_{i_1}, \cdots, a_{i_N})$, $\mathrm{i} = 1, \cdots, n^N$; $i_1, \cdots, i_N = 1, \cdots, \mathrm{n}$. The correlation among the symbols in a N-dimensional message can be represented by their joint probabilities $p(\alpha_i) = p(a_{i_1} a_{i_2} \cdots a_{i_N})$.
  - The correlation among the symbols can also be represented by their conditional probabilities. For instance, if the source sends a symbol sequence $\{a_{i_1}, \cdots, a_{i_m}\}$, the probability of the source sending the symbol $a_{i_{m+1}}$ at the moment i_{m+1} is $p(a_{i_{m+1}}/a_{i_1}, \cdots, a_{i_m})$. A symbol sent by the source is only correlated with the previous m symbols. This is a Markov info source having a memory length m.

2D Discrete Memory Info. Source

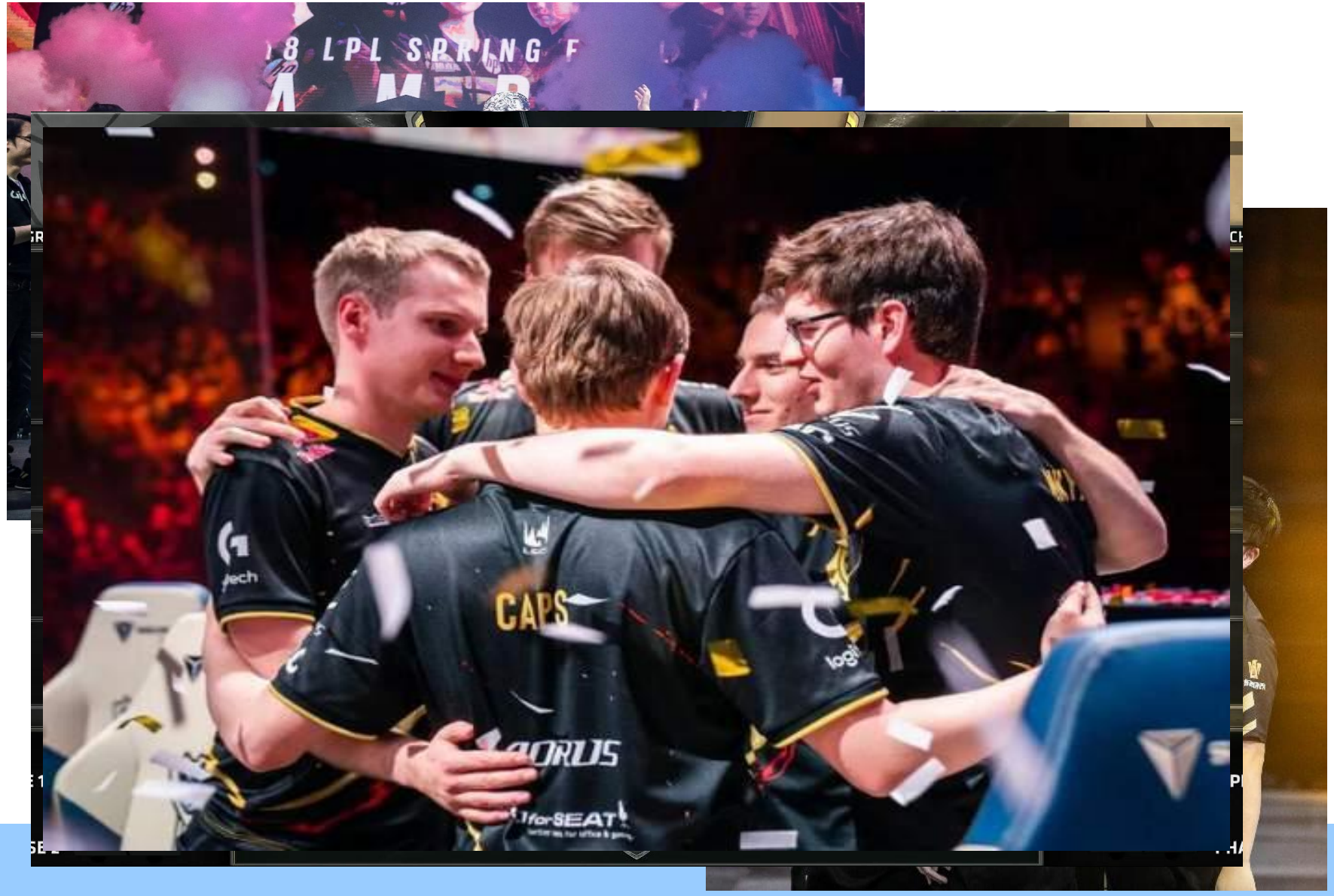
- Assume that a pair of single-symbol discrete info source X_1 and X_2 send the symbols chosen from $\{a_1, \cdots, a_n\}$, the resultant two-dimensional discrete stationary memory info source $\boldsymbol{X} = X_1 X_2$ send a two-symbol message chosen from

$$\boldsymbol{X} \in \{a_1 a_1, \cdots, a_1 a_n, \cdots, a_n a_1, \cdots, a_n a_n\}.$$

This message space is constituted by all the possible combinations of two symbols chosen from the space $\{a_1, \cdots, a_n\}$. A two-symbol message can be represented by $\alpha_i = (a_{i_1} a_{i_2})$, where $i_1, i_2 = 1,2, \cdots, n$. The mathematical model of $\boldsymbol{X}$ is

$$\begin{pmatrix} \boldsymbol{X} \\ P(\boldsymbol{X}) \end{pmatrix} = \begin{pmatrix} \alpha_1, & \alpha_2, & \cdots & \alpha_{n^2} \\ p(\alpha_1), & p(\alpha_2), & \cdots & p(\alpha_{n^2}) \end{pmatrix},$$

where $\{p(\alpha_i) = p(a_{i_1} a_{i_2}) | 1 \le i \le n^2, i_1, i_2 = 1,2, \cdots, n\}$. These probabilities constitute a complete probabilistic space.


LPL SPRING
CAPS

Properties of 2D Info. Source

- The entropy of the 2D info source $\boldsymbol{X}$ is calculated as

$$H(\boldsymbol{X}) = H(X_1X_2) = H(X_1) + H(X_2/X_1)$$

- When the random symbols X_1 and X_2 are independent of each other，the entropy of the 2D info source $\boldsymbol{X}$ can be simplified as

$$H(\boldsymbol{X}) = H(X_1X_2) = H(X_1) + H(X_2)$$

- When X_1 and X_2 are identically and independently distributed, the 2D info source $\boldsymbol{X}$ can be simplified as the expanded source of $X_1 = X_2 = X$ having an order of 2. Its entropy is expressed as

$$H(\boldsymbol{X}) = H(X^2) = 2H(X)$$

- The entropy of the 2D memory info source is not higher than the sum of the corresponding memoryless source, namely

$$H(X_1X_2) \leq H(X_1) + H(X_2)$$

Review

- Given two random information source X and Y, their joint entropy, their individual entropies and their conditional entropies satisfy the following equations:

$$\mathrm{H}(XY) = H(X) + H(Y/X)$$

$$\mathrm{H}(XY) = H(Y) + H(X/Y)$$

This is the additive properties of the entropy.

Proof: Exploit the definitions of the entropy, conditional entropy and joint entropy, as well as the probabilistic properties of the random variables to prove the additive property.

- When information sources X and Y are independent of each other, their joint entropy can be the summation of their individual entropies:

$$\mathrm{H}(XY) = H(X) + H(Y)$$

N-D Discrete Memory Info. Source

- The entropy of the N-D discrete memory info source $\boldsymbol{X}$ (**chain rule**):

$$H(\boldsymbol{X}) = H(X_1) + H(X_2/X_1) + \cdots + H(X_N/X_1X_2\cdots X_{N-1})$$

Chain rule: the entropy of the N-D discrete memory info source X is the sum of the conditional entropies having different orders.

- The conditional entropy of the the N-D discrete memory info source $\boldsymbol{X}$ is a non-increasing function of the dimension N, namely

$$H(X_N/X_1X_2\cdots X_{N-1}) \leq H(X_N/X_2X_3\cdots X_{N-1})$$

Stationary of the source $= H(X_{N-1}/X_1X_2\cdots X_{N-2})$

When $N = 3$, we have $H(X_3/X_1X_2) \leq H(X_3/X_2) = H(X_2/X_1)$.

Generally, the non-increasing can be represented as:

$$H(X_N/X_1X_2\cdots X_{N-1}) \leq H(X_{N-1}/X_1X_2\cdots X_{N-2})$$
$$\leq \cdots \leq H(X_2/X_1) \leq H(X_1)$$

Entropy Rate and Entropy limit

- The entropy $H(\boldsymbol{X}) = H(X_1 X_2 \cdots X_N)$ of the N-D discrete info source quantify the information carried by a message having N symbols. The average information carried by a single symbol sent by the source $\boldsymbol{X}$ is

$$H_N(\boldsymbol{X}) = \frac{1}{N} H(X_1 X_2 \cdots X_N),$$

  This is the **entropy rate** of the N-D info source. When the dimension N tends to infinity, we obtain **the entropy limit of a single symbol**:

$$H_\infty(\boldsymbol{X}) = \lim_{N\to\infty} \frac{1}{N} H(X_1 X_2 \cdots X_N)\,.$$

- The correlation among symbols sent by a practical info source is not constrained in N dimensions but in infinite dimensions. The entropy limit accurately quantifies the information carried by a single symbol. When the source is stationary, the entropy limit is

$$H_\infty(\boldsymbol{X}) = \lim_{N\to\infty} H_N(\boldsymbol{X}) = \lim_{N\to\infty} H(X_N / X_1 X_2 \cdots X_{N-1})\,.$$

Example

- We have a binary information source X sending a message sequence consisting of symbols 0 and 1. This is memoryless source, since it always sends the symbol 1 with a probability of $p(1) = 0.6$ and it sends the symbol 0 with a probability of $p(0) = 0.4$, which is not correlated to the previous symbols.
  - Is this information source stationary?
  - Please calculate $H(X^2), H(X_3/X_1X_2)$ and $H_\infty(X)$。
  - Please calculate $H(X^4)$ and write down all the possible messages sent by the information source X^4.